

MANISH MAN SHRESTHA

Firmware & Embedded Systems Engineer | IoT Developer

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Hiring Team

Senior Embedded Firmware Engineer Search - Mahwah, NJ

Re: Senior Embedded Firmware Engineer (Contract, Mahwah, NJ)

Dear Hiring Manager,

I am excited to apply for the Senior Embedded Firmware Engineer contract role. Over 15 years I have delivered embedded firmware for connected products from concept through production, with deep, hands-on strength in exactly the areas this role emphasizes: low-level firmware, board bring-up, wireless connectivity, and embedded architecture.

Here is how my experience maps to the role:

- **C/C++ on 32-bit ARM with RTOS:** I build real-time firmware in Embedded C/C++ across ARM Cortex-M (STM32, NXP/LPC) and ESP32 - device drivers, ISR design, and FreeRTOS task scheduling under hard timing constraints.
- **Board bring-up and low-level debugging:** I routinely perform hardware bring-up and peripheral integration (UART, SPI, I2C, RS-485, ADC, GPIO) and debug with oscilloscopes, logic analyzers, and JTAG/SWD.
- **Wireless, TCP/IP, and connected products:** Wi-Fi/BLE, secure MQTT/TLS telemetry, and lwIP TCP/IP stacks; I built distributed ESP32-S3 IoT nodes validated over a 120-day field deployment with zero failures, optimizing power consumption and footprint.
- **Your preferred list is my core:** ESP-IDF / ESP32 modules, bootloaders and OTA-safe / secure firmware updates, a custom I2C SSD1306 OLED display driver, Azure IoT cloud connectivity, and Docker-based CI/CD pipelines for repeatable firmware builds.
- **Architecture, testing, and mentoring:** I contribute to embedded architecture and design reviews, build structured unit/integration test and validation workflows, and have mentored engineers as both a professor and a CTO.

This role is an unusually close fit for my background, and I would welcome the chance to contribute from day one. I am glad to work on-site in Mahwah, NJ. Thank you for your consideration.

Sincerely,

Manish Man Shrestha